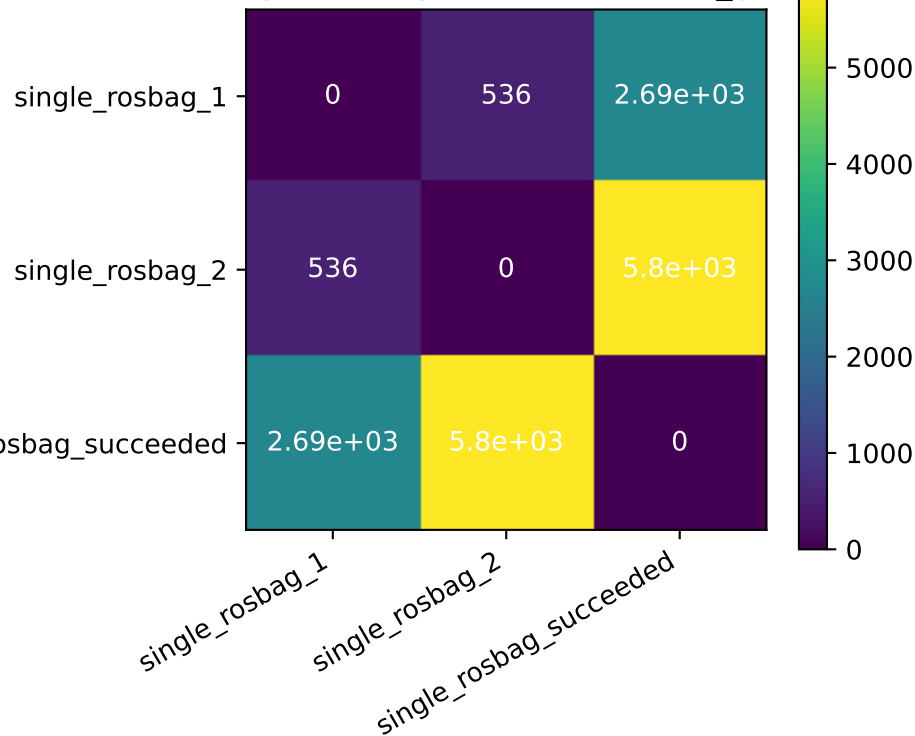
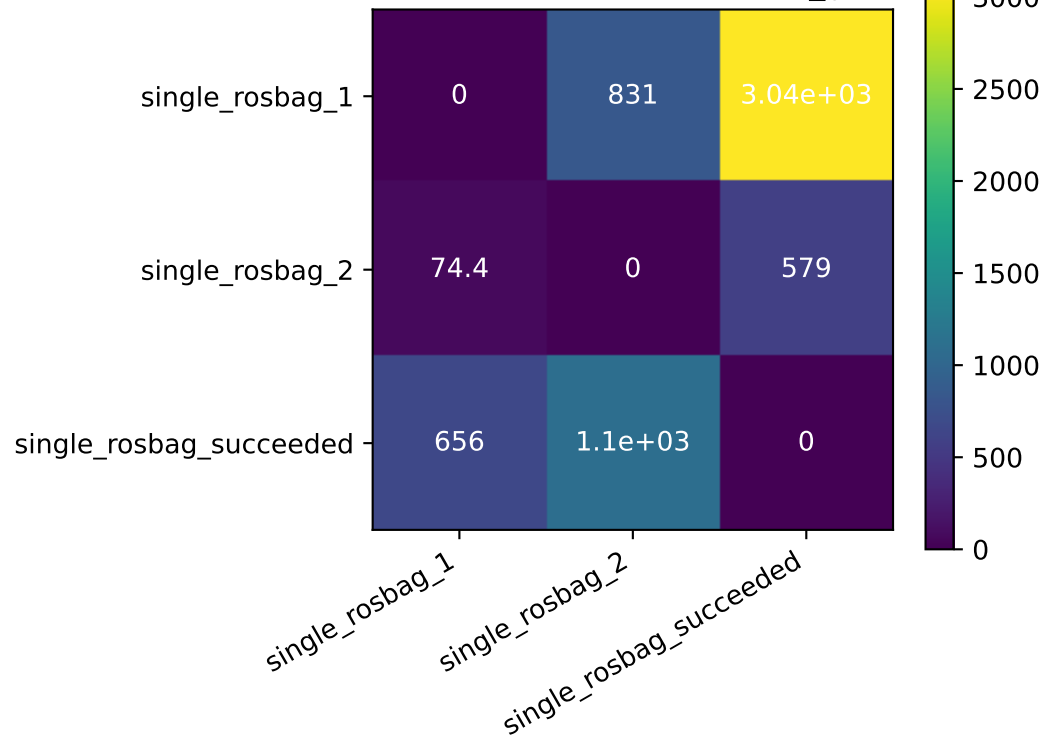


Three-bag scale-quotient consensus

pairwise quotient distance d_{ij}



cross-evaluation Delta L_{ij}



Three-bag physical and cross-evaluation audit

Gauge-invariant physical quantities

single_rosbag_1

```
J/m = [[ 1.015805e-01 -1.228804e-04 -1.649156e-02]
[-1.228804e-04  7.735542e-02  9.522580e-06]
[-1.649156e-02  9.522580e-06  1.715480e-01]]
f/m = [0.334909 0.431263 0.376598 0.312238]
c    = [ 0.017628 -0.004117 -0.044376]
cross costs = [ 268.528887 1099.385419 3305.442753]
```

single_rosbag_2

```
J/m = [[ 0.073724  0.001825 -0.00043 ]
[ 0.001825  0.034796 -0.010867]
[-0.00043  -0.010867  0.105213]]
f/m = [0.381802 0.309009 0.34086  0.409455]
c    = [-0.026435  0.011339  0.009415]
cross costs = [187.059741 112.652483 691.905564]
```

single_rosbag_succeeded

```
J/m = [[ 0.231928 -0.005147  0.018047]
[-0.005147  0.10967  -0.003059]
[ 0.018047 -0.003059  0.128895]]
f/m = [0.444531 0.435005 0.229507 0.27868 ]
c    = [-0.019823 -0.055131 -0.005031]
cross costs = [ 684.664834 1130.166667  28.342745]
```